Università degli Studi di Padova

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE



Digital Forensics & Biometrics

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Lecture 06 – Network forensics

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IDS TRIGGER

IDS are merely packet sniffers: need to teach the IDS what is an attack.



Anomaly Detection (Profile based)

analyzes computer activity and network traffic looking for anomalies.

Anomaly: deviation from what is defined normal (by system administrator) -> *user group profile*
e.g. you won't expect telnet towards a SMTP server

How to build a user group profile
statistical sampling, rule-based, and neural networks.

Pros:

- Intruders do not know if they generate alarms (attack can not be configured on signatures)
- Detect internal attacks
- Profiles are dynamic

Cons:

- High initial prep time
- No protection during training
- Constant update as users' habits change
- Defining normal behavior can be difficult
- False positives, false negatives
- Hard to understand

Misuse Detection (Signature based)

Use signature files to detect intrusive activity (analogous to signature files commonly used in virus-scanning)

Pros:

- Files created on known attack (if matching, the attack is sure)
- Given the files, protection is instantaneous
- Easier to understand and configure (manager knows what is going on)

Cons:

- Inability to detect new or unknown attacks
- Inability to detect variations of known attacks
- Signature database administration
- Sensors must maintain state information

TRACEBACK STRATEGIES

Network forensics deals with capture, recording, analysis and investigation of network traffic to traceback the attackers.

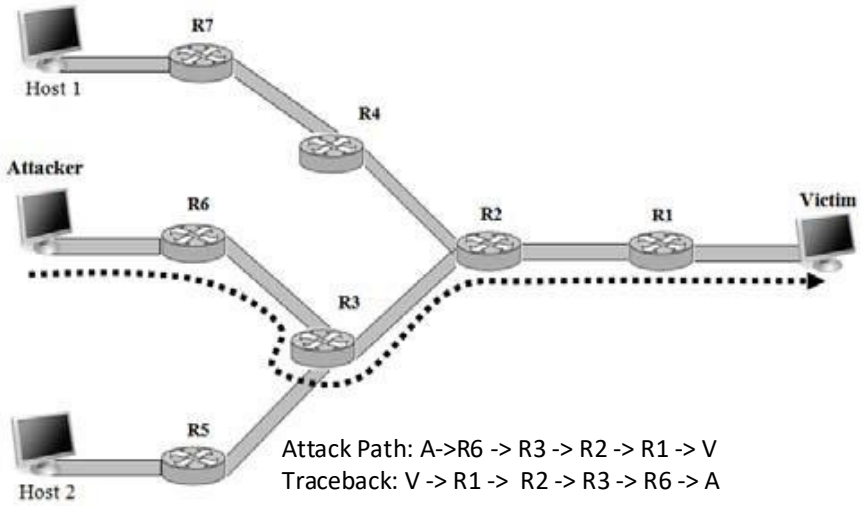
Problem: given a packet or a file, find the originating terminal.

Used in DDOS attack or againts IP spoofing

Characteristics:

Time of analysis (*real time/post mortem*)

Data source (*flow based/packet based*)



Assumptions:

- attackers are able to generate and send any packet
- multiple attackers may act coordinated
- attackers are aware of the traceback ability
- routers have limited processing and storage capabilities
- routers are rarely compromised and all routers may not participate in traceback
- suspicious packet stream are made of a few packets

Requirements:

- compatibility with existing network protocols, routers and infrastructure
- simple and minimal number of functions on transit routers
- support for incremental implementation, partial deployment and scalability
- minimal overhead of time and resources (processing, bandwidth, memory)
- fast convergence involving a few packets
- minimal involvement of the internet service provider (ISP)

TRACEBACK SOLUTIONS

Logging packets at key routers and later mining them for attack path reconstruction

source path isolation engine (SPIE): hash of multiple fields in packet
autonomous management network (AMN): monitoring manager receives requests from sensors

Packet-marking places part or the complete address of the router into the IP packet randomly with a fixed probability or only once deterministically

probabilistic packet marking (PPM)
advanced and authenticated packet marking (AAPM): 8-bit hash of address
algebraic packet marking (APM): algebraic techniques to calculate 15-bit marks as points on polynomials.
fast internet traceback (FIT): uses a fragment of the hash, the number of fragment and a distance field
deterministic packet marking (DPM)
deterministic edge router marking (DERM)

Hybrid traceback approaches integrate packet marking and packet logging to achieve the advantages of both the techniques.

distributed link list traceback (DLLT): router marks a packet, stores current IP address and packet ID in the marking table
hierarchical IP traceback system (HITS)
hybrid single packet IP traceback (HIT)
logging and deterministic packet marking (LDPM)

Autonomous System can be a group of networks regulated by one or more entity, which enforces a clearly defined routing policy.

Traceback strategies are defined here as well.

ATTACK RECONSTRUCTION

In order to refine defenses, it is necessary to analyze and understand how attackers works.

Log files are extremely important!!



Honeypot refers to a set of services, an entire operating system or even an entire network that is built to lure and contain intruders

- Designed to be compromised
- Well monitored
- No essential activity is carried on
- A way to draw attack away from more crucial elements in a network
- Useful tool to study attackers.

Problems:

From a legal point of view, honeypots can be problematic.

- honeypot has no value, and therefore, it is not possible to legally claim any damages.
- borderline between keeping attackers out of a network and inviting them.
- may be challenged as an unfair entrapment.

PRIVACY-PRESERVING NETWORKS

DEEP WEB FORENSICS

- Through Deep Web, criminal activities can be carried in anonymous way using bitcoins in transactions
- Silk Road bust – oct. 2013: $34.5M bitcoins seized by FBI, $1.5B estimated worth of 1.5M transactions.
- $300-400K/day: drug market sales
- Dark Web vs. DeepWeb: 25% sites of Deep Web involved in illegal activities
  - Drug, illegal material exchange (child pornography, terrorism propaganda, industrial data)
  - Identity Theft
  - Counterfeit money



Problems:

Internet size: 3.17B users worldwide

Anonimity: Criminal are anonymous, Bitcoins allow untraceable transactions, finding culpables can be hard

Jurisdiction: locations can be difficult to find, transnational rules make legal actions more complex

Solutions:

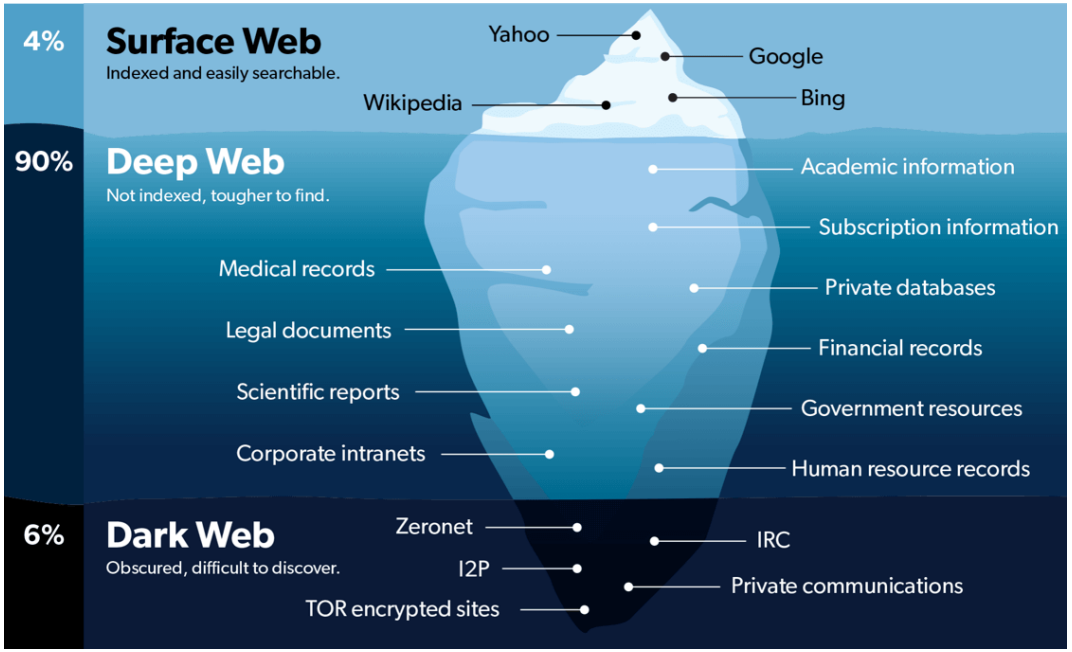
Training: several agencies (Interpol, FBI, police corpse) are investing money

Data collection methods: methods to navigate dark web, developing malware to compromise servers, hidden service/consumer data/social sites monitoring, marketplace profiling, semantic analysis

Human errors: investigators sometimes rely on the fact that usernames or contacts (and communication ways) are revealed

DEEP WEB APPLICATIONS

- Virtual Private Network (VPN) with TOR
  See next slides
- Free Anonymous Internet (FAI)
  used in blockchain because it is a decentralized deep web
- FreeNet
  Network of peers to peers and designed to securely and privately spread information on the Internet.
- ZeroNet
  torrent with Bitcoin encryption.



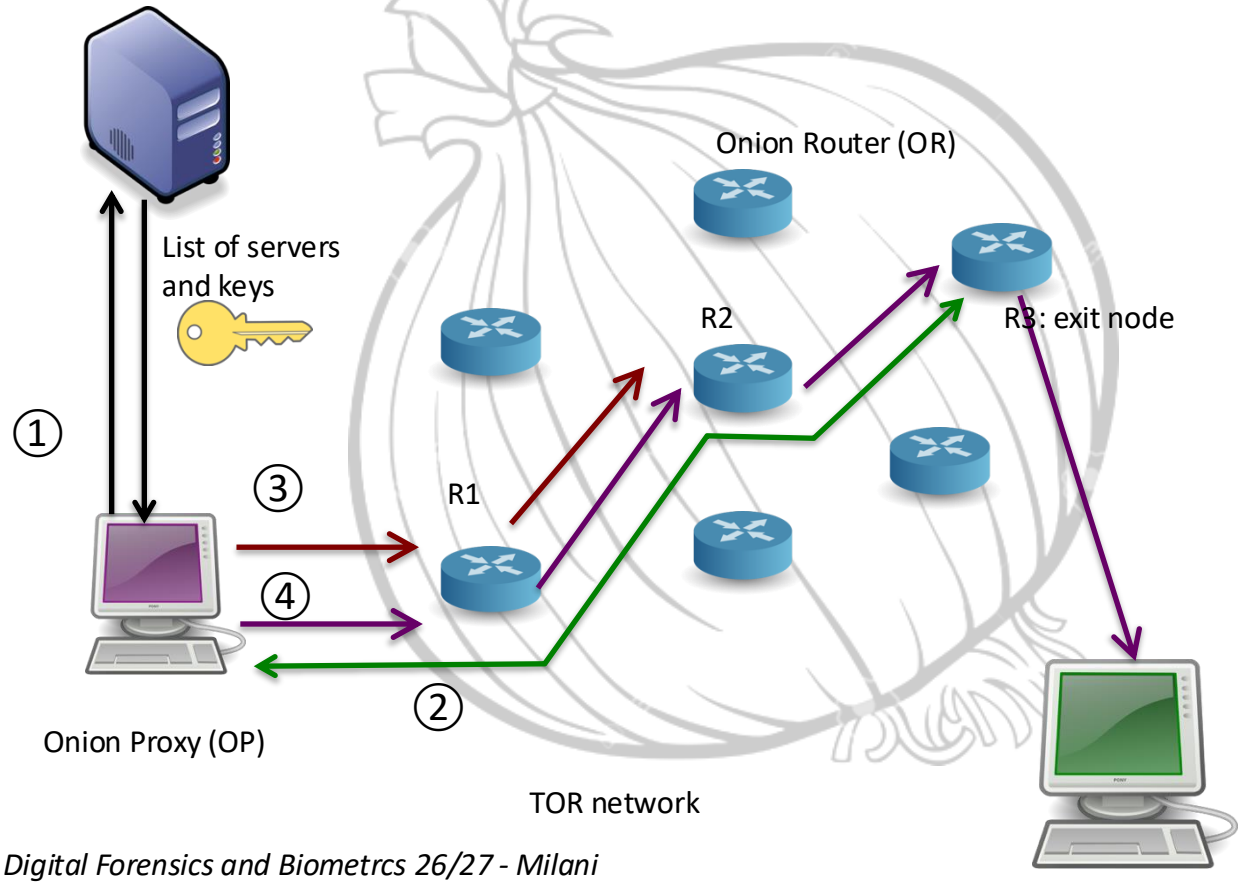
TOR NETWORK

R. Dingledine and N. Matheson, 2002
based on Onion Routing (mid '90s, Paul Syverson, M. G. Reed e D. Goldschlag)

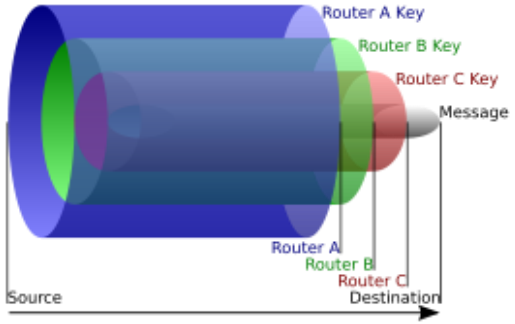
https://www.torproject.org

Protect privacy and anonimity.

- Data are encrypted multiple times (nested encryption)
- Every router correspond to an encryption layer
- No router knows the destination and the sending nodes.



Once connection is established, data are exchanged in cells of 512 bytes (to avoid profiling)



Connection has been created: all messages will be sent as K1(K2(K3(DATA)))

TOR FORENSICS

Advantages:

- Everything is encrypted
- No-one knows both sending and destination nodes
- Packet lengths are disguised

Weaknesses:

- Timing analysis
- Weakness in the exit node

The secrets of the ToR browser

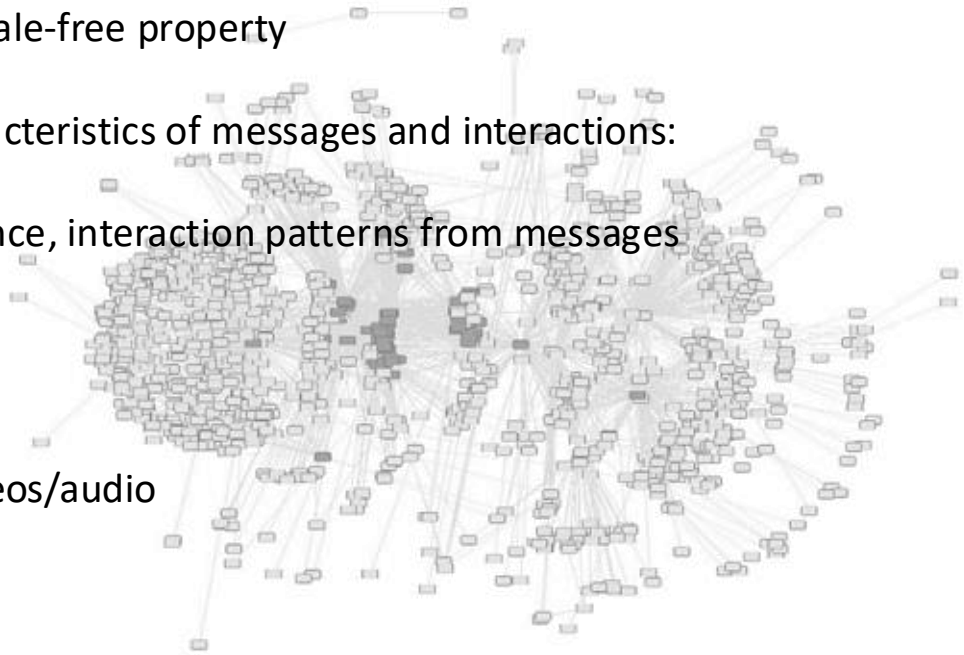


Joint analysis of RAM and disks + network streams

Strategy	solutions	Aims
RAM analysis	Belkasoft RAM capturer Hex dump Volatility Recall	• Dumps • Artifacts in RAM (URLs, .onion links, emails, search terms, data) • Network traces (connections and IPs of relays)
File system and storage (registers)	Pecmd, Regshot	• Prefetch • TOR installation and windows registers • Configuration folders, db files
Network analysis	Wireshark Onion sniffer netstat	• Port (9150, 9151) • Packet lengths and timestamp • connections

OTHER SOLUTIONS

- **Spidering**
  - dark web crawler that collect information on web sites (*focused* on a specific topic, *hidden:* automatically filling hidden forms)
  - Features: URL, BoW characterizing documents,
- **Link and content analysis**
  - Measure similarity between contents
  - Methodologies: information sources, domain spidering, backlink search, filtering, indexing and analysis, clustering, activity modelling
- **Dark Network Analysis**
  - Network analysis: topology
  - Graph modelling, small-world property, scale-free property
- **Interactional coherence analysis**
  - Check communication and verify the characteristics of messages and interactions: interaction patterns
  - Lexical relations, direct address, co-reference, interaction patterns from messages
- **Authorship analysis**
  - Writing style features
  - Image/multimedia data analysis
- **Sentiment analysis**
  - Semantic classification of text/images/videos/audio
- **Affect analysis**



COMBINE EVIDENCES

COMBINING EVIDENCES

Multiple measures need to be combined. Different evidence combination strategies, e.g. Bayesian

Core idea:

H: hypotheses (e.g., computer C is infected)
E: evidence (e.g., system calls)
P(**H**): prior probability before seeing the evidence
P(**E** | **H**): probability of seeing evidence E is H is true (likelihood)
P(**H** | **E**): posterior probability of the hypothesis being true

$$P(H|E) = \frac{P(H) \cdot P(E|H)}{P(E)}$$

As for multiple evidences (e.g., multiple IDS or values), independent

$$P(\boldsymbol{H}|\boldsymbol{E_1}, \boldsymbol{E_2}, \dots, \boldsymbol{E_n}) \propto P(H) \cdot P(\boldsymbol{H}|\boldsymbol{E_1}, \boldsymbol{E_2}, \dots, \boldsymbol{E_n}) = P(H) \cdot \prod_i P(\boldsymbol{H}|\boldsymbol{E_i})$$

e.g.,

$\boldsymbol{E_1}$: login timestamp.
$\boldsymbol{E_2}$: IP address.
$\boldsymbol{E_3}$: browser artifact.
….

Main issues:

- probabilities that are hard to estimate reliably.
- Independence assumption

EXAMPLE

H: PC-17 is infected. Prior=5%

Evidence	*Observation*
E_1	firewall logs show `PC-17` contacted suspicious IP `185.x.x.x`
E_2	DNS logs show `PC-17` resolved a known malware domain
E_3	proxy logs show download of an executable from that domain

Assume $P[E_1|H] = 0.9$ but $P[E_1|\neg H] = 0.1$. (there can be contact but no infection)

$$P[H|E_1] = \frac{P[E_1|H] \cdot P[H]}{P[E_1|H] \cdot P[H] + P[E_1|\neg H] \cdot P[\neg H]} = \frac{0.9 \cdot 0.05}{0.9 \cdot 0.05 + 0.1 \cdot 0.95} = 0.321$$

Adding DNS evidence $P[E_2|H] = 0.85$ but $P[E_2|\neg H] = 0.05$

$$P[H|E_1, E_2] = \frac{0.85 \cdot 0.321}{0.85 \cdot 0.321 + 0.679 \cdot 0.05]} = 0.889$$

Adding proxy. $P[E_3|H] = 0.8$ but $P[E_3|\neg H] = 0.08$
I get

$$P[H|E_1, E_2, E_3] = 0.98$$

Alternatively, multiply likelihood ratios

$$LR_i = \frac{P(E_i|H)}{P(E_i|\neg H)}$$

COMBINING EVIDENCES

4 people (Barbara, John, Sean and Ken) are friends living in the same apartment.

- One morning, Ken is found dead, stabbed with a knife.
- Not suicide (stabbed in the back)
- No-one entered the house.
- Assume only one killer.



Witness 1: "I am 80% sure I heard men arguing last night"

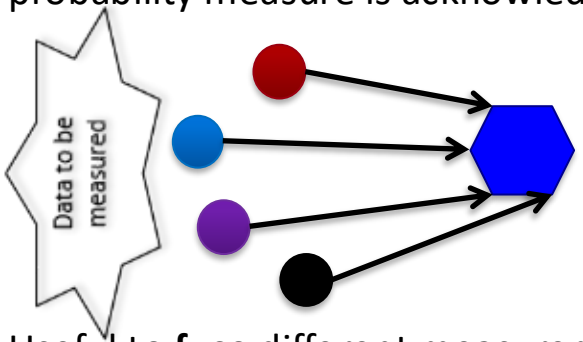
Witness 2: "I am 60% sure that Ken was going to lend some money to Sean; no reason for him to kill."

DEMPSTER-SCHAFER THEORY

introduced by A. P. Dempster in 1967, later developed by his student G. Schafer (1976)

"general framework for reasoning with uncertainty, with understood connections to other frameworks such as probability, possibility and imprecise probability theories" from Wikipedia

probability measure is acknowledged, but not precisely known



Also known as **theory of belief functions** or **theory of evidence**

Multiple (heterogeneous) evidences -> one belief

Useful to **fuse** different measurements (see *sensor fusion*)

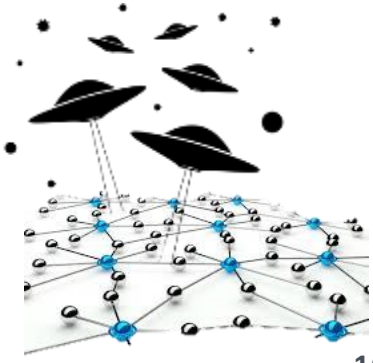
Each sensor has a *degree of support for the fact*: 0 = no support 1 = full support

e.g. F={network is under attack!}

Differently from Bayesian approach:

- Beliefs and their negation do not need to sum to 1
  $B[F] + B[\overline{F}] \neq 1$ but $P[F] + P[\overline{F}] = 1$

- $B[F] = B[\overline{F}] = 0$ is possible (no evidence for or against the fact)



DS THEORY: DEFINITIONS

Let X be the *universe* (set of possible states/conclusions for the analyzed system)

$A \subseteq X$ is:

- mutually exclusive (at most one is true)
- exhaustive (at least, one is true)

The power set Θ is the set of all possible subsets of X including $\rightarrow$

e.g. $X = \{a, b\}$ $\Theta = \{\rightarrow, \{a\}, \{b\}, X\}$ *subset = hypothesis*

Frame of discernment

Bayes was concerned with evidence that supported single conclusions
(e.g., evidence E for each outcome A in X s.t. $P[A|F]$)

DS theory is concerned with evidences that support subsets of outcomes $A \cup B \cup C$ *or* $\{A, B, C\}$
i.e. uncertainty between groups of outcomes

Comments:

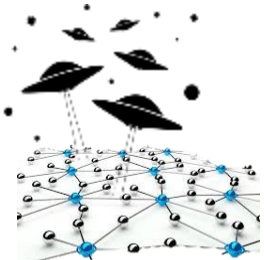
- $P[\rightarrow] = 0$ at least one of the options must be true
- $0 \leq P[A] \leq 1$
- $P[X] = 1$ one outcome must be true

Example: Network attack case.

$X = \{attack, normal\}$

$\Theta = \{\rightarrow, \{attack\}, \{normal\}, \{normal, attack\}\}$

uncertain



TRANSFERABLE BELIEF MODEL (TBM)

Basic belief assignment (BBA): $m(A): 2^X \to [0,1]$ **Mass function**

Proportion of evidences that support this set

Two properties:

① $m(\emptyset) = 0$

② $\sum_{A \subseteq 2^X} m(A) = 1$

The mass m (A) of A expresses the proportion of all relevant and available evidence that supports the claim that the actual state belongs to A but to no particular subset of A (they have their own mass).

Proportion of evidences that support this set

E.g. $m(A \cup B) = 0.3$ means *"there is evidence for {A v B } that can not be divided into evidences for A or B"*

Belief (or support) function

$$bel(A) = \sum_{B:\ B \subseteq A} m(B)$$

Plausibility function

$$pl(A) = \sum_{B:\ B \cap A \neq \emptyset} m(B)$$

Note:

$$bel(A) \leq P[A] \leq pl(A)$$

It is possible to state that

$$pl(A) = 1 - bel(\bar{A})$$

Disbelief or dis(A)

Mobius transform

$$m(A) = \sum_{B:\ B \subseteq A} -1^{|A|-|B|} bel(B)$$

where |A | – | B| is the difference of cardinalities

INTERPRETATION OF MASS

Random switch model
Think of the evidence as a switch that oscillates randomly between two or more positions. The function m represents the fraction of the time spent by the switch in each position.

Voting model
m represents the proportion of votes cast for each of several results of the evidence (possibly including that the evidence is inconclusive).

Envelope model
Think of the evidence as a sealed envelope, and m as a probability distribution on what the contents may be

INTERPRETATION OF BELIEF AND PLAUSIBILITY

Belief (or possibility) is the probability that A is provable (supported) by the evidence.

Plausibility is the probability that A is compatible with the available evidence (cannot be disproved).

Belief interval $[bel(A) \;\; pl(A)]$

Useful to decide whether you need more evidences

[0,1] : total ignorance
[0,0] : belief that A is false
[1,1] : belief that A is true
[0.3,1] : partial belief that A is true
[0,0.8] : partial disbelief that A is true
[0.2,0.7] : partial belief and disbelief that A is true

Probability is related to this $bel(A) \leq P[A] \leq pl(A)$

MORE THAN 1000 WORDS ... EXAMPLES

4 people (Barbara, John, Sean and Ken) are friends living in the same apartment.

- One morning, Ken is found dead, stabbed with a knife.
- Not suicide (stabbed in the back)
- No-one entered the house.
- Assume only one killer.





$$X = \{B, J, S\} \qquad \Theta = \{\rightarrow, \{B\}, \{J\}, \{S\}, \{B,J\}, \{B,S\}, \{J,S\}, \{B,J,S\}\}$$

After analyzing the crime scene, detectives assign mass functions



Event	Set	Mass	bel	pl
No-one is guilty	$\rightarrow$	0		
B is guilty	{B}	0.1	0.1	0.4
J is guilty	{J}	0.2	0.2	0.7
S is guilty	{S}	0.1	0.1	0.6
Either B or J is guilty	{B,J}	0.1	0.4	0.9
Either B or S is guilty	{B,S}	0.1	0.3	0.8
Either J or S is guilty	{J,S}	0.3	0.6	0.9
One of the three is guilty	{B,J,S}	0.1	1	1

John owed money to Ken.

Witnesses who heard people arguing from outside thought they were men

DS'S COMBINATION RULE

Two different source of evidences

Assuming the conditional independence, $P[s_1, s_2 \mid d] = P[s_1 \mid d] + P[s_2 \mid d]$

① Generate a belief value from the two measurements

$$m(C) = \sum_{A_i \cap B_j = C} m_1(A_i)\, m_2(B_j)$$

② Normalize with $K = \sum_{A_i \cap B_j \to} m_1(A_i)\, m_2(B_j)$

Beliefs can be combined in order to obtain a higher accuracy.

EXAMPLE: CRIME SCENE

Witness 1:

"I am 80% sure I heard men arguing last night"

Θ	mass
J,S	0.8
B,J,S	0.2





Witness 2: "I am 60% sure that Ken was going to lend some money to Sean; no reason for him to kill."

Θ	mass
B,J	0.6
B,J,S	0.4

It is possible to combine the belief together in order to make a decision



$$m_{1,2}(J) = m_1(\{J,S\}) \square m_2(\{B,J\}) = 0.8 \square 0.6 = 0.48$$

$$m_{1,2}(\{J,S\}) = m_1(\{J,S\}) \square m_2(\{B,J,S\}) = 0.8 \square 0.4 = 0.32$$

$$m_{1,2}(\{B,J\}) = m_1(\{B,J,S\}) \square m_2(\{B,J\}) = 0.2 \square 0.6 = 0.12$$

$$m_{1,2}(\{B,J,S\}) = m_1(\{B,J,S\}) \square m_2(\{B,J,S\}) = 0.2 \square 0.4 = 0.08$$

Possibilities are now expanded! Stronger belief against John!

DS is a clear generalization of the Bayesian theory that avoids the problem of assigning non-available prior probabilities and makes no assumptions about non-available probabilities.

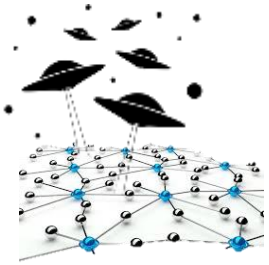
EXAMPLE: FUSION OF MULTIPLE IDS

Network attack case.

$$X = \{attack, normal\}$$

$$2^X = \{\rightarrow, \{attack\}, \{normal\}, \{normal, attack\}\}$$

uncertain



Sensor A

2^X	mass
attack	0.32
normal	0.25
uncertain	0.43

Sensor B

2^X	mass
attack	0.35
normal	0.1
uncertain	0.55

Product table

	{A} 0.32	{N} 0.25	{A,N} 0.43
{A} 0.35	0.11 ★	0.09	0.15 ★
{N} 0.1	0.03	0.025 △	0.04 △
{A,N} 0.55	0.18 ★	0.14 △	0.24 ■

★ Bel(A)= 0.11+0.18+0.15 = 0.44 = 0.5

△ Bel(N)= 0.025+0.04+0.14 = 0.205 = 0.23

■ Bel({A,N})= 0.24 = 0.24 = 0.27

normalize

EXAMPLE: FUSION OF MORE IDS

		Sensor					
		#1	*#2*	*#3*	*#4*	*#5*	*#6*
Hypothesis	*Normal*	0.3	0.217	0.667	0.667	0.217	0.217
	Attack	0.4	0.567	0.167	0.167	0.567	0.567
	Uncertainty	0.3	0.216	0.166	0.166	0.216	0.216

		Iteration				
		#1 - #2	*R - #3*	*R - #4*	*R - #4*	*Final Results*
Hypothesis	*Normal*	0.262	0.857	0.187	0.746	0.475
	Attack	0.65	0.107	0.751	0.247	0.524
	Uncertainty	0.088	0.036	0.062	0.007	0.001

PROS AND CONS OF DS THEORY

Pros

- High potential for managing uncertainty
- Assigns probability to uncertainty
- Does not require a priori knowledge
- Suitable for detecting previously unknown attacks

Cons

- Computation complexity increases exponentially with the number of possible event outcomes
- Conflicting beliefs management assigning empty belief value
- Evidences should be completely independent

EXERCISE A.6

Given a network which is monitored by a three IDS with beliefs

Sensor A

Decision	mass
Attack	0.3
Safe	0.2
Don't know	0.5

Sensor B

Decision	mass
Attack	0.35
Safe	0.25
Don't know	0.4

Sensor C

Decision	mass
Attack	0.35
Safe	0.1
Don't know	0.55



what is the likelihood that the network is under attack?

EXERCISE A.5

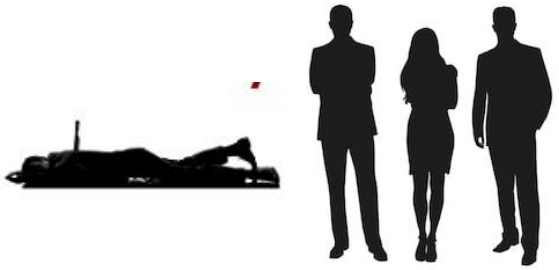
Given a system failure, two experts (John and Paul) are evaluating the different HIDS and NIDS of different machines to identify the source of the attack. System failure could be caused be Terminal A, Terminal B or Terminal C.

John believes that the failure is due to terminal A with a mass m1(A) = 0.8 or to terminal B with mass m1(B) = 0.1; uncertainty is 0.1.

Paul believes that the failure is due to computer C with m2(C) = 0.9 or to the others with mass 0.05 (m2(A,B)); uncertainty is 0.05.

Compute a Dempster-Schafer combination of John and Paul opinions.
Describe each step and make a table of values.

CONTRADICTIONS IN DS THEORY



Witness 1:

⊖	mass
J	0.99
S	0.01
B	0.0

Witness 2:

⊖	mass
J	0.0
S	0.01
B	0.99



W1+ W2	mass
J	0.99 x 0.0 = 0.0 | 0.00
S	0.01 x 0.01 = 0.0001 | **1.00**
B	0.0 x 0.99 = 0.0 | 0.00